PROPYLTHIOURACIL

1st trimester & thyroid storm

Severe hepatotoxicity

CHEMICAL NAME 6-propyl-2-sulfanylidene-2,3-dihydropyrimidin-4(1H)-one
CLASS Thionamide antithyroid drug
TRADE NAMES PTU · Propycil · Propylthiouracil (generic)

THERAPEUTIC USES / INDICATIONS

- Hyperthyroidism in the first trimester of pregnancy.
- Thyroid storm (also blocks peripheral T4→T3 conversion).
- Patients intolerant of carbimazole/thiamazole (not agranulocytosis).

ADMINISTRATION

- **PO** in divided doses (every 6–8 h) due to short half-life.
- Thyroid storm: may give via NG tube or rectally.

DOSAGES

- **Hyperthyroidism:** 300–600 mg/day initially (single or divided doses) until euthyroid; maintenance 50–150 mg/day.
- **Thyrotoxic crisis:** 200 mg every 4–6 h on the first day, reducing as control is gained (UK label).
- Switch to thiamazole after first trimester.

PHARMACOKINETICS

- **Onset:** 1–2 weeks clinically. Highly protein-bound (less placental transfer).
- Metabolised in liver.
- Excreted in urine. **t½:** 1–2 h.

PHARMACODYNAMICS

- Inhibits **thyroid peroxidase** and peripheral 5′-deiodinase (↓ T4 → T3).

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Rash, pruritus.
- Nausea, taste loss.
- Arthralgia.

ADVERSE EFFECTS (SERIOUS)

- Severe hepatotoxicity / acute liver failure (boxed warning; more than other thionamides).
- Agranulocytosis.
- ANCA vasculitis.

CONTRAINDICATIONS & CAUTIONS

- Hepatic impairment.
- Previous severe reaction.
- Not first-line outside pregnancy/storm.

KEY INTERACTIONS

- **Warfarin:** altered effect.

NURSING MANAGEMENT

- Baseline and regular LFTs; report fatigue, anorexia, jaundice.
- FBC if sore throat/fever.
- Thyroid storm: monitor temperature, HR, BP, mental status; β-blocker, steroids, iodine as ordered (iodine ≥1 h after PTU).

PATIENT EDUCATION

- Report tiredness, loss of appetite, yellow skin, dark urine, sore throat or fever immediately.
- Expect a switch of medicine after the first trimester.

KEY POINTS

- Thionamide + blocks T4→T3 conversion.
- Use in 1st trimester and thyroid storm.
- Severe hepatotoxicity.
- Short t½ — divided doses.

SOURCES CHECKED **[635]** Propylthiouracil 100mg Tablets (UK SmPC) **[636]** Propylthiouracil, oral (US label) **[637]** Propylthiouracil, oral (US label, American Health Packaging)
Full references p. 432–457